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Indian Standard

IS 1656 : 2022

शिशु आहार — दुग्ध-अनाज आधारित
अनुपूरक आहार — विशिष्टि
(पांचवा पुनरीक्षण)

Infant Food — Milk-Cereal Based
Complementary Foods —
Specification
(*Fifth Revision*)

ICS 67.100.99

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Price Group 7

Dairy Products and Equipment Sectional Committee, FAD 19

FOREWORD

This Indian Standard (Fifth Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Dairy Products and Equipment Sectional Committee had been approved by the Food and Agriculture Division Council.

Milk-cereal based complementary foods are mainly intended to accustom the infant's digestive tract to solid foods. Nutritionally, this category of foods serves as an important source of calories to meet the energy requirements due to increased physical activity of infant. Beginning with the introduction of cereals after six months of age, the intake is slowly increased so that by 8 months of age and onwards, nearly half of the total intake is from milk and remaining from cereals and a variety of other foods. With such a feeding practice, milk almost completely meets the protein requirements, while cereals and/or other foods meet the energy and satiety requirements of the infant.

Under the *Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act*, 1992, various types of foods for infants being marketed in our country have been placed under the following two categories:

- a) Infant milk substitutes, and
- b) Infant foods.

'Infant milk substitute' means any food being marketed or otherwise represented as partial or total replacement for mother's milk, whereas 'Infant Food' means any food being marketed or otherwise represented as a complement to mother's milk to meet the growing nutritional needs of the infant after the age of six months and up to the age of two years.

This standard covers the requirements for milk-cereal based complementary foods and a separate standard, namely, IS 11536 : 2022 'Processed cereal based complementary foods for infants — Specification (*third revision*)' exists which covers the requirements for processed cereal based complementary foods.

This standard was first published in 1960 and subsequently revised in 1969, 1985, 1997 and then in 2007 to update the chemical and microbiological requirements and recommendatory list of sources of vitamin compounds and minerals salts; and to include list of permitted food additives.

A scheme for labelling environment friendly products known as ECO-Mark was also introduced in the fourth revision at the instance of the Ministry of Environment and Forests, Government of India. The ECO-Mark shall be administered by the Bureau of Indian Standards (BIS) under the *BIS Act*, 2016 as per the Resolution No. 71 dated 20 February 1991 and No. 425 dated 28 October 1992 published in the Gazette of the Government of India. For a product to be eligible for marking with the ECO-Mark, it shall also carry the Standard Mark of BIS for quality besides meeting additional environment friendly (EF) requirements given in the standard, which are based on the Gazette Notification No. GSR 624 (E) dated 6 September 1995 for labelling beverages, infant foods and processed fruits and vegetable products as Environment Friendly Products, published in the Gazette of the Government of India.

Amendments were later issued to IS 1656 : 2007 in November 2012 and November 2014.

This fifth revision has been brought out to harmonize the requirements of milk-cereal based complementary foods with the *Food Safety and Standards (Foods for Infant Nutrition) Regulations*, 2020. Major changes include:

- a) Product description has been updated;
- b) List of permitted optional ingredients has been updated;
- c) List of permitted food additives as well as their levels have been updated;
- d) Referred test methods have been updated;
- e) Chemical and microbiological requirements have been updated; and
- f) Labelling clause has been modified.

(Continued on third cover)

Indian Standard

**INFANT FOOD — MILK-CEREAL BASED
COMPLEMENTARY FOODS — SPECIFICATION**

(Fifth Revision)

1 SCOPE

This standard prescribes the requirements, methods of test and sampling for milk-cereal based complementary foods intended to complement the diet of infants after six months up to twenty four months of age.

2 REFERENCES

The standards listed in Annex A contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Milk-Cereal Based Complementary Food — Milk-cereal based complementary foods commonly called as weaning foods or supplementary foods means the food for infants after 6 months up to 24 months of age, which is based on milk (*see* 4 of IS 13688), cereals and/or legumes (pulses), millets, nuts and defatted edible oilseed extracts and so prepared as to permit dilution with water or milk or other suitable medium. It may also contain soybean, fruits and vegetables or their products, egg or egg products, protein concentrates or protein isolates or protein hydrolysates, milk solids, various carbohydrates, and salts such as phosphates and citrates.

3.2 Routine Tests — The tests carried out on each lot to check the essential requirements which are likely to vary during production.

3.3 Type Test — The tests to prove conformity to the requirements of this standard. These are intended to approve the formulation and quality of the product at least in the beginning of marketing or certification or both. These tests are also conducted periodically to supplement the routine tests or whenever the basic formula or method is changed.

4 REQUIREMENTS

4.1 Description

The milk-cereal based complementary food shall be in the form of powder, small granules or flakes, free from lumps and shall be uniform in appearance. It shall also be free from dirt, and extraneous matter, preservatives, added colour and flavour and from any material which are harmful to infant's health. The flavour and odour of the milk-cereal based complementary food in the powder form or when reconstituted with water shall be fresh and sweet (*see* IS 10641). It shall be free from rancid taste and musty odour (*see* IS 10641).

4.2 The product shall be reasonably free from scorched particles.

4.3 Lactose and glucose polymers shall be the preferred carbohydrates for milk-cereal based complementary food. Sucrose and/or fructose shall not be added, unless needed as a carbohydrate source, and provided the sum of these does not exceed 20 per cent of total carbohydrate.

NOTE — For the purpose of the above clause, record of the addition may be maintained by the manufacturer since analytical methods have not been established for these.

4.4 The milk-cereal based complementary food shall not contain hydrogenated vegetable fats and oils.

4.5 Optional Ingredients

The milk-cereal based complementary food may contain in addition to the vitamin and minerals listed under Table 2, other nutrients ordinarily found in human milk in amounts prescribed in Table 1 (either per 100 g or 100 kcal basis).

4.5.1 The product may also contain algal and fungal oil as sources of docosahexaenoic acid (DHA) and arachidonic acid (ARA) from *Cryptocodinium cohnii*, *Mortierella alpina*, *Schizochytrium* sp., and *Ulkenia* sp. or fish oil at the level of maximum 0.5 percent DHA of total fatty acids and ratio of ARA : DHA as 1 : 1 minimum.

NOTE — DHA content shall not be less than 0.2 per cent of total fatty acids, if a claim related to the addition of DHA is made.

IS 1656 : 2022

Table 1 Optional Ingredients
(Clause 4.5)

Sl No. (1)	Nutrient (2)	Requirements per 100 g (3)	Requirements per 100 kcal (4)	Method of Test, Ref to (5)
i)	Pantothenic acid, mg	2.00 - 10.00	0.40 - 2.10	IS 16642
ii)	Vitamin B ₁₂ , µg	0.25 - 0.70	0.05 - 0.15	IS 16640* or AOAC 2014.02
iii)	Vitamin K, µg	7.50 - 19.00	1.60 - 4.00	IS 21446
iv)	Choline, mg, <i>Max</i>	32.00	6.80	IS 17668
v)	Inositol, mg	20.00 - 200.00	4.25 - 42.55	IS 16649* or any other validated international method
vi)	Calcium, mg	250.00 - 700.00	53.20 - 148.90	IS 12760 or ISO 15151 or ISO 21424*
vii)	Phosphorus, mg	125.00 - 500.00	26.60 - 106.40	IS 12756 or ISO 15151 or ISO 21424*
viii)	Chloride, mg	250.00 - 800.00	53.20 - 170.20	IS 11763 or AOAC 2016.03*
ix)	Magnesium, mg	30.00 - 75.20	6.40 - 16.00	IS 12760 or ISO 15151 or ISO 21424*
x)	Sodium, mg	90.00 - 300.00	19.15 - 63.80	IS 12760 or ISO 15151 or ISO 21424*
xi)	Selenium, µg	5.00 - 40.00	1.00 - 9.0	IS 15303 or ISO 15151 or ISO 20649*
xii)	Taurine, mg, <i>Max</i>	60.00	12.75	AOAC 2018.06* or AOAC 997.05
xiii)	L-amino acids, mg, <i>Min</i>	0.90	0.20	AOAC 2018.06
xiv)	L-Carnitine, mg, <i>Min</i>	5.00	1.00	IS 17668
xv)	Biotin, µg	7.50 - 50.00	1.60 - 10.60	IS 17670
xvi)	Iodine, µg	90.00 - 225.60	19.15 - 48.00	IS 17379
xvii)	Potassium, mg	300.00 - 900.00	63.80 - 191.50	IS 12760 or ISO 15151 or ISO 21424*
xviii)	Pyridoxine, µg	100.00 - 400.00	21.30 - 85.10	IS 17669*

NOTES:

1 When any of the nutrients listed in this table are added, the amount of these added nutrients shall be declared on the label. A variation of minus 10.0 per cent from the declared value of the nutrients or nutritional ingredients on the label shall be allowed.

2 In case of dispute, the method indicated by '*' shall be the referee method.

4.6 Food Additives

for infants ready for consumption prepared following manufacturer's instructions, unless otherwise indicated:

4.6.1 Following additives may be used in the preparation of milk-cereal based complementary food

Food Additives	Recommended Maximum Level per 100 g of the Product Ready for Consumption
Emulsifiers	
Lecithins	1 500 mg
Mono- and diglycerides	5 000 mg singly or in combination ¹
Acetic fatty acid esters of glycerol	
Lactic fatty acid esters of glycerol	
Citric acid fatty acid esters of glycerol	
Acidity regulators	
Sodium hydrogen carbonate	GMP
Calcium carbonate	
Potassium hydrogen carbonate	
L(+)-lactic acid	
Citric acid	
Acetic acid	
Potassium acetate	
Sodium acetate	
Calcium acetate	
Malic acid (DL)-L(+) form only	
Sodium lactate (solution)-L(+) form only	
Potassium lactate (solution)-L(+) form only	
Calcium lactate-L(+) form only	
Monosodium citrate	
Trisodium citrate	
Monopotassium citrate	
Tripotassium citrate	
Calcium citrate	
Hydrochloric acid (food grade)	
Sodium hydroxide	
Potassium hydroxide	
Calcium hydroxide	
Glucono-delta-lactone	
L(+)-Tartaric acid	500 mg singly or in combination
Disodium tartrate	
Potassium sodium L(+) tartrate L(+) form only	
Ortho phosphoric acid	Only for pH adjustment, 440 mg singly or in combination as phosphorous
Mono sodium ortho phosphate	
Disodium orthophosphate	
Trisodium orthophosphate	
Monopotassium orthophosphate	
Dipotassium orthophosphate	
Tripotassium orthophosphate	
Monocalcium orthophosphate	
Dicalcium orthophosphate	
Tricalcium orthophosphate	

IS 1656 : 2022

Food Additives	Recommended Maximum Level per 100 g of the Product Ready for Consumption
Antioxidants	
Mixed tocopherols concentrate	300 mg/kg fat or oil, singly or in combination
Alpha tocopherol	
L-Ascorbyl palmitate	200 mg/kg fat or oil
L-ascorbic acid	50 mg expressed as ascorbic acid
Sodium ascorbate	
Potassium ascorbate	
Calcium ascorbate	20 mg expressed as ascorbic acid
Raising Agents	
Ammonium carbonate	GMP
Ammonium hydrogen carbonate	
Sodium carbonate	
Sodium hydrogen carbonate	
Thickeners	
Carob bean gum	1 000 mg singly or in combination; 2 000 mg in gluten free cereal based foods
Guar gum	
Gum Arabic	
Xanthan gum	
Pectins (amidated and non-amidated)	
Oxidized starch	5 000 mg singly or in combination
Mono starch phosphate	
Distarch phosphate	
Phosphated distarch phosphate	
Acetylated distarch phosphate	
Acetylated distarch adipate	
Starch acetate esterified with acetic anhydride	
Starch sodium octenyl succinate	
Acetylated oxidized starch	
Anticaking Agent	
Silicon dioxide	GMP
Packaging gases	
Carbon dioxide	GMP
Nitrogen	
Enzymes	
Alpha amylase	GMP

4.6.2 For reasons of stability and safe handling, some vitamins and other nutrients have to be converted into suitable preparations. For this purpose, the following food additives may be used as nutrient carriers in milk-cereal based complementary food:

Additive/Carrier	Recommended Maximum Level in Ready-to-Use foods for Infant Nutrition (mg/kg)
Gum arabic (gum acacia)	10
Silicon dioxide	10
Mannitol (for vitamin B ₁₂ dry rubbing, 0.1 percent only)	10
Starch sodium octenyl succinate	100
Sodium L-ascorbate (in coating of nutrient preparations containing polyunsaturated fatty acids)	75

4.7 Quality of Ingredients

4.7.1 All ingredients used shall be clean, of good quality, safe and suitable for ingestion by infants.

4.7.2 The vitamins, minerals and other nutrients shall be of food grade. Wherever applicable, milk-cereal based complementary foods shall use the source compounds for minerals, vitamins and other nutrients as follows:

a) Minerals

- 1) *Calcium (Ca)* — Calcium carbonate, calcium chloride, calcium citrate (tricalcium di citrate), calcium gluconate, calcium glycerophosphate, calcium L-lactate, calcium hydroxide, calcium phosphate monobasic (calcium dihydrogen phosphate), calcium phosphate dibasic (calcium hydrogen phosphate), calcium phosphate tribasic (tricalcium diphosphate), calcium oxide;
- 2) *Phosphorous (P)* — Calcium phosphate monobasic, calcium phosphate dibasic, calcium phosphate tribasic, magnesium phosphate dibasic, magnesium phosphate tribasic, potassium phosphate monobasic, potassium phosphate dibasic, sodium phosphate dibasic, Phosphoric acid;
- 3) *Chloride (Cl)* — Calcium chloride, choline chloride, magnesium chloride, manganese chloride, potassium chloride, sodium chloride, hydrochloric acid (food grade);
- 4) *Iron (Fe)* — Ferrous carbonate (stabilized with saccharose), ferrous citrate, ferrous

fumarate, ferrous gluconate, ferrous succinate, ferrous lactate, ferric ammonium citrate, ferric citrate, ferrous bisglycinate, sodium ferric pyrophosphate ferric diphosphate, ferric orthophosphate, ferrous sulphate, hydrogen reduced iron, electrolytic iron, carbonyl iron, ferric saccharate, sodium ferric diphosphate;

- 5) *Magnesium (Mg)* — Magnesium hydroxide carbonate, magnesium chloride, magnesium oxide, magnesium phosphate dibasic (magnesium hydrogen phosphate), magnesium phosphate tribasic (trimagnesium phosphate), magnesium carbonate, magnesium sulphate, magnesium hydroxide, magnesium salts of citric acid, magnesium gluconate, magnesium lactate, magnesium glycerol-phosphate, magnesium acetate;
- 6) *Sodium (Na)* — Sodium bicarbonate, sodium carbonate, sodium chloride, trisodium citrate, sodium gluconate, sodium L-lactate, sodium phosphate monobasic (sodium dihydrogen phosphate), sodium phosphate dibasic (disodium hydrogen phosphate), sodium phosphate tribasic (trisodium phosphate), sodium sulphate, sodium hydroxide;
- 7) *Potassium (K)* — Potassium bicarbonate, potassium carbonate, potassium chloride, potassium citrate (tripotassium citrate), potassium gluconate, potassium L-lactate;
- 8) *Copper (Cu)* — Copper gluconate (cupric gluconate), cupric carbonate, cupric citrate, copper sulphate (cupric sulphate);
- 9) *Iodine (I)* — Potassium iodide, sodium iodide, potassium iodate;
- 10) *Zinc (Zn)* — Zinc acetate, zinc chloride, zinc oxide, zinc sulphate, zinc gluconate, zinc lactate, zinc citrate (zinc citrate dihydrate or zinc citrate trihydrate);
- 11) *Manganese (Mn)* — Manganese(II) carbonate, manganese(II) chloride, manganese(II) citrate, manganese sulphate, manganese (II) gluconate, manganese (II) glycerol-phosphate; and
- 12) *Selenium* — Sodium selenate, sodium selenite, sodium hydrogen selenite.

b) Vitamins

- 1) *Vitamin A* — Retinyl acetate, retinyl palmitate, trans retinol;
- 2) *Provitamin A* — Beta-carotene;
- 3) *Vitamin D* — Vitamin D₂ (ergocalciferol), vitamin D₃ (cholecalciferol);
- 4) *Vitamin E* — D-alpha-tocopherol, DL-alpha-tocopherol, D-alpha-tocopheryl acetate, DL-alpha-tocopheryl acetate;

IS 1656 : 2022

- 5) *Vitamin K₁* — Phytomenadione (2-Methyl-3-phytyl-1,4-naphthoquinone/Phylloquinone/phytonadione);
 - 6) *Vitamin K₂* — Menaquinone;
 - 7) *Thiamin (Vitamin B₁)* — Thiamin chloride hydrochloride, thiamin mononitrate;
 - 8) *Riboflavin (Vitamin B₂)* — Riboflavin, riboflavin-5-phosphate sodium;
 - 9) *Niacin* — Nicotinamide, nicotinic acid;
 - 10) *Pantothenic Acid* — Calcium -D- pantothenate, D-panthenol, sodium-D-pantothenate, DL-panthenol;
 - 11) *Vitamin B₆* — Pyridoxine hydrochloride; Pyridoxal-5-phosphate;
 - 12) *Folic Acid* — N-Pteroyl-L-glutamic acid;
 - 13) *Biotin (Vitamin H)* — D-biotin;
 - 14) *Vitamin B₁₂* — Cyanocobalamin, hydroxocobalamin; and
 - 15) *Vitamin C* — L-ascorbic acid, sodium-L-ascorbate, calcium-L-ascorbate, potassium-L-ascorbate, 6-palmitoyl-L-ascorbic acid (ascorbyl palmitate).
- c) Amino Acids
- 1) *L-Arginine*;
 - 2) *L-Arginine hydrochloride*;
 - 3) *L-Cystine*;
 - 4) *L-Cystine dihydrochloride*;
 - 5) *L-Cysteine*;
 - 6) *L-Cysteine hydrochloride*;
 - 7) *L-Histidine*;
 - 8) *L-Histidine hydrochloride*;
 - 9) *L-Isoleucine*;
 - 10) *L-Isoleucine hydrochloride*;
 - 11) *L-Leucine*;
 - 12) *L-Leucine hydrochloride*;
 - 13) *L-Lysine*;
 - 14) *L-Lysine mono hydrochloride*;
 - 15) *L-Methionine*;
 - 16) *L-Phenylalanine*;
 - 17) *L-Threonine*;
 - 18) *L-Tryptophan*;
 - 19) *L-Tyrosine*; and
 - 20) *L-Valine*.
- d) Carnitine
- 1) *L-Carnitine*; and
 - 2) *L-Carnitine hydrochloride*.
- e) Choline
- 1) *Choline*;
 - 2) *Choline chloride*;
 - 3) *Choline citrate*;
 - 4) *Choline hydrogen tartrate*; and
 - 5) *Choline bitartrate*.
- f) Inositols
- 1) *Myo-inositol*.

4.8 The product shall also conform to the requirements given in Table 2 (either per 100 g or 100 kcal basis) and Table 3.

4.9 If the product claims to have higher quality protein, the protein efficiency ratio (PER), when tested as per IS 7481 shall be minimum 2.5.

Table 2 Requirements for Milk-Cereal Based Complementary Foods

(Clause 4.8)

Sl No.	Characteristic	Requirement per 100 g	Requirements per 100 kcal	Method of Test, Ref to
(1)	(2)	(3)	(4)	(5)
i)	Moisture, percent by mass, <i>Max</i>	5.00	—	IS 11623 for reference purpose IS 16072 for routine purpose
ii)	Protein, percent by mass			IS 11917
	a) Total Protein including milk protein, (N × 6.25), <i>Min</i>	15.0	3.20	
	b) Milk protein (N × 6.38), <i>Min</i>	5.0	1.00	
iii)	Fat, percent by mass			Annex B
	a) Total fat, including milk fat, <i>Min</i>	7.5	1.60	
	b) Milk fat ¹⁾ , <i>Min</i>	5.0	1.00	
iv)	Total carbohydrates, per cent by weight, <i>Min</i>	55.0	11.70	Annex C
v)	Total ash, per cent by weight, <i>Max</i>	5.0	—	Annex C of IS 14433
vi)	Acid insoluble ash in dilute hydrochloric acid, per cent by weight, <i>Max</i>	0.10	—	Annex D of IS 14433

Table 2 (Concluded)

Sl No.	Characteristic	Requirement per 100 g	Requirements per 100 kcal	Method of Test, Ref to
(1)	(2)	(3)	(4)	(5)
vii)	Vitamin A (as retinol equivalent RE), µg	350.00 - 823.00	75.00 - 175.00	IS 16639
viii)	Vitamin C, mg	25.00 - 75.00	5.30 - 16.00	IS 5838 or IS 17176*
ix)	Iron, mg	3.00 - 7.00	0.60 - 1.50	AOAC 985.35 or ISO 15151 or ISO 21424*
x)	Crude fibre (on dry basis), percent by mass, <i>Max</i>	1.00	—	IS 10226 (Part 1)
xi)	Vitamin D, µg	5.00 - 14.00	1.00 - 3.00	IS 17177
xii)	Thiamine, mg	0.20 - 0.50	0.04 - 0.10	IS 17669
xiii)	Riboflavin mg	0.40 - 2.00	0.08 - 0.40	IS 17669
xiv)	Niacin equivalent, mg	3.80 - 9.90	0.80 - 2.10	IS 17669
xv)	Dietary Folate equivalent (DFE) ²⁾ , µg	15.00 - 50.00	3.20 - 10.60	AOAC 2013.13
xvi)	Zinc, mg	2.50 - 5.90	0.50 - 1.25	15 of IS 1699 or ISO 15151 or ISO 21424*

NOTES:

1 In case of dispute, the method indicated by ‘*’ shall be the referee method.

2 For the purpose of Type tests, all tests mentioned above are to be carried out and for the purpose of Routine tests, the tests given from Sl. No. (i) to (ix) are to be carried out.

3 A variation of minus 10.0 per cent from the declared value of the nutrients or nutritional ingredients on the label shall be allowed. The nutrient levels shall not exceed maximum limits as specified in Table 2.

¹⁾ Since there is no reliable methods at present for the estimation of separate contents of milk fat and vegetable fat in the total fat, records of their addition shall be maintained by the manufacturer. However, the product shall not have less than 7.5 percent total fat when tested as per Annex B.

²⁾ 1 microgram DFE = 0.6 microgram folic acid.

Table 3 Microbiological Requirements for Milk-Cereal based Complementary Food
(Clause 4.8)

SI No.	Characteristic	Requirement				Method of Test, Ref to
		Sampling Plan		Limit (cfu)		
		n	c	m	M	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
(i)	Aerobic plate count	5	2	1×10^3 /g	1×10^4 /g	IS 5402 (Part 1)
(ii)	Coliform count	10	0	< 10/g	—	IS 5401 (Part 1)
(iii)	<i>Staphylococcus aureus</i> (Coagulase positive)	5	0	< 10/g	—	IS 5887 (Part 8/Sec 1* or 2)
(iv)	Yeast and mould count	5	0	< 10/g	—	IS 5403 or IS 16069-2*
(v)	<i>Salmonella sp.</i>	15	0	Absent/25 g	—	IS 5887 (Part 3/Sec 1)
(vi)	<i>Listeria monocytogenes</i>	10	0	Absent/25 g	—	IS 14988 (Part 1)
(vii)	<i>Bacillus cereus</i>	5	2	1×10^2 /g	5×10^2 /g	IS 5887 (Part 6)
(viii)	<i>Sulphite reducing clostridia</i>	5	2	10/g	1×10^2 /g	ISO 15213
(ix)	<i>E. coli</i>	10	0	Absent/g	—	IS 5887 (Part 1) or IS 16424*

NOTES:

1 For sampling plan, see Annex D.

2 In case of dispute, the method indicated by ‘*’ shall be the referee method.

3 The requirement for *Salmonella* shall be tested in a laboratory situated away from the production area.

IS 1656 : 2022

4.10 The pesticide residues, antibiotic and veterinary drug residues, naturally occurring toxins and other contaminants, if any, in the raw materials used in the manufacture of the product shall not exceed the limits as prescribed in the *Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011*.

4.11 Heavy metals and other contaminants or toxic substances (melamine), if any, in the product shall not exceed the limits specified in Table 4.

Table 4 Limits of Heavy Metals and Other contaminants or Toxic Substances

(Clause 4.11)

Sl No.	Contaminant/Toxin	Limit	Method of Test,Ref to
(1)	(2)	(3)	(4)
i)	Lead, ppm, <i>Max</i>	0.2	IS 12074 or AOAC 2015.01*
ii)	Copper, ppm	2.8 to 15.0	15 of IS 1699 or ISO 15151 or ISO 21424*
iii)	Arsenic, ppm, <i>Max</i>	0.05	IS 11124 or AOAC 2015.01
iv)	Tin, ppm, <i>Max</i>	5.0	ISO 14377
v)	Cadmium, ppm, <i>Max</i>	0.1	15 of IS 1699 or AOAC 2015.01*
	Melamine, ppm, <i>Max</i>	2.5	IS 16195

NOTE — In case of dispute, the method indicated by ‘*’ shall be the referee method.

4.12 Hygienic Conditions

The product shall be processed, packed, stored and distributed under strict hygienic conditions as prescribed in IS 2491.

4.13 Optional Requirements for Eco-Mark

4.13.1 General Requirements

4.13.1.1 The product shall conform to the requirements prescribed under 4.1 to 4.12.

4.13.1.2 The manufacturers shall produce the consent clearance as per the provisions of *Water (PCP) Act, 1974*, *Water (PCP) Cess Act, 1977* and *Air (PCP) Act, 1981* along with the authorization if required under *Environment (Protection) Act, 1986* and the Rules made thereunder to the Bureau of Indian Standards, while applying for the ECO-Mark and the product shall also be in accordance with the *Food Safety and Standards Act, 2006* and the Rules made thereunder. Additionally, the *Legal Metrology (Packaged Commodities) Rules, 2011* have to be complied with.

4.13.1.3 The product/packaging may also display in brief the criteria based on which the product has been labelled environment friendly.

4.13.1.4 The material used for product packing shall be recyclable or biodegradable.

4.13.1.5 The product shall be microbiologically safe when tested as per IS 5887 (Part 5) and should be free from bacterial and fungal toxins.

4.13.1.6 The product package or leaflet accompanying it may display instruction of proper use, storage and transport (including refrigeration temperature compliance) so as to maximize the product performance, safety and minimize wastage.

4.13.2 Specific Requirements

4.13.2.1 The material used inside tile metal cap of the product shall conform to the relevant Indian Standards of food grade plastics as permitted under the *Food Safety and Standards Act, 2006* and the Rules and Regulations made there under. Caps and closures shall not be treated as labels.

4.13.2.2 No synthetic food colour and artificial sweetener shall be added or used in the product.

4.13.2.3 The percentage of fruit juice/pulp, if any added shall be mentioned on the product package.

5 PACKING AND MARKING

5.1 Packing

5.1.1 The milk-cereal based complementary foods shall be packed in hermetically sealed, clean and sound metal containers (*see* IS 11078) or in a flexible pack so as to protect it from deterioration. In case plastic material is used for flexible packaging, only food grade plastic shall be used (*see* IS 10171). The packaging material used for product shall be free from Bisphenol A (BPA).

5.1.2 The product shall be packed in quantities as stipulated under *Legal Metrology (Packaged Commodities) Rules, 2011* as well as in accordance with requirements under the *Food Safety and Standards (Packaging) Regulations, 2018*.

5.2 Marking

5.2.1 The containers shall bear legibly and indelibly with the following information:

- Name of the product, and brand name, if any;
- Name and address of the manufacturer;
- Batch or lot or code number;
- Month and year of manufacturing or packing;
- Net quantity;

- f) Date before which the contents should be consumed be indicated by marking the words 'Use by date/recommended last consumption date/Expiry date(month and year)';
- g) Directions for storage;
- h) Composition — Indicating the approximate composition of nutrients per 100 g or per 100 ml of the product as well as the energy value in kilocalories or kilojoules;
- j) In case fish oil is used, it shall be mentioned on the label;
- k) Statement in bold letters: "IMPORTANT NOTICE : MOTHER'S MILK IS BEST FOR YOUR BABY";
- m) A statement that infant food shall be used only on the advice of a health worker as to the need for its use and the proper method of its use;
- n) A warning that infant food is not the sole source of nourishment of an infant;
- p) A statement indicating instruction for appropriate and hygienic preparation including cleaning of utensils, bottles and teats and warning against health hazards of inappropriate preparations;
- q) Feed chart and directions for use;
- r) Instructions for discarding leftover feed;
- s) Instructions for use of measuring scoop (level or heaped) and the quantity per scoop (scoop to be given with pack); and
- t) Any other requirements as stipulated under *Food Safety and Standards (Labelling and Display)*

Regulations, 2020; Infant Milk Substitutes, Feeding Bottles and Infant Foods Act, 1992 and Rules 1993; and Legal Metrology (Packaged Commodities) Rules, 2011.

5.2.1.1 In case of flexible packs, a cautionary notice to the following effect shall be printed on the container: 'On opening, transfer the contents of the pack to a clean air tight container. After each use, replace the lid tightly and store in a cool dry place.'

5.2.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

5.2.3 ECO-Mark

The Product may also be marked with the ECO-Mark, the details of which may be obtained from the Bureau of Indian Standards.

6 SAMPLING

Representative samples of the material shall be drawn and tested for conformity to this standard as prescribed in Annex F of IS 14433.

NOTE — The crude fibre content [Table 2, SI No. (x)] shall be tested on the composite sample.

IS 1656 : 2022

ANNEX A

(Clause 2)

LIST OF REFERRED STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
1070 : 1992	Reagent grade water — Specification (<i>third revision</i>)	(Part 6) : 2012/ ISO 7932 : 2004	Horizontal method for the enumeration of presumptive <i>Bacillus cereus</i> — Colony count technique at 30°C (<i>first revision</i>)
1699 : 1995	Methods of sampling and test for food colours (<i>second revision</i>)		
2491 : 2013	Food hygiene — General principles — Code of practice (<i>third revision</i>)	(Part 8/ Sec 1) : 2002/ ISO 6888-1:1999	Horizontal method for enumeration of coagulase-positive <i>Staphylococci</i> (<i>Staphylococcus aureus</i> and other species), Section 1 Technique using Baird-Parker agar medium
5401 (Part 1) : 2012/ ISO 4832 : 2006	Microbiology of food and animal feeding stuffs — Horizontal method for the detection and enumeration of coliforms: Part 1 Colony count technique (<i>second revision</i>)	(Part 8/ Sec 2) : 2002/ ISO 6888-2 : 1999	Horizontal method for enumeration of coagulase-positive <i>Staphylococci</i> (<i>Staphylococcus aureus</i> and other species), Section 2 Technique using rabbit plasma fibrinogen agar medium
5402 (Part 1) : 2021	Microbiology of the food chain — Horizontal method for the enumeration of microorganisms: Part 1 Colony count at 30 C by the pour plate technique	7481 : 1974	Method for determination of protein efficiency ratio (PER)
5403 : 1999	Method for yeast and mould count of foodstuffs and animal feeds (<i>first revision</i>)	10171 : 1999	Guide on suitability of plastics for food packaging (<i>second revision</i>)
5838 : 1970	Methods for estimation of vitamin C in foodstuffs	10226 (Part 1) : 1982/ ISO 5498 : 1981	Method for determination of crude fibre content in food products: Part 1 General method
5887	Methods for detection of bacteria responsible for food poisoning	10641 : 1983	Recommended methods for determination of aroma and taste thresholds
(Part 1) : 1976	Isolation, identification and enumeration of <i>Escherichia coli</i> (<i>first revision</i>)	11078 : 2012	Round open top sanitary cans for milk powder — Specification (<i>second revision</i>)
(Part 3/Sec 1) : 2020/ ISO 6579-1 : 2017	Horizontal method for the detection, enumeration and serotyping of <i>Salmonella</i> , Section 1 Detection of <i>Salmonella</i> spp. (<i>third revision</i>)	11124 : 1984	Method for atomic absorption spectrophotometric determination of arsenic
(Part 5) : 1976	Isolation, identification and enumeration of <i>Vibrio cholerae</i> and <i>Vibrio parahaemolyticus</i> (<i>first revision</i>)	11623 : 2008/ ISO 5537 : 2004	Dried milk — Determination of moisture content (Reference Method) (<i>first revision</i>)

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
11721 : 2013/ ISO 1736 : 2008	Dried milk and dried milk products — Determination of fat content — Gravimetric method (Reference method) (<i>second revision</i>)	14988 (Part 1) : 2020/ ISO 11290-1: 2017	Microbiology of the food chain — Horizontal method for detection and enumeration of <i>Listeria monocytogenes</i> and of <i>Listeria</i> spp.: Part 1 Detection method (<i>first revision</i>)
11763 : 2011/ ISO 5943 : 2006	Cheese and processed cheese products — Determination of chloride content — Potentiometric titration method (<i>third revision</i>)	15303 : 2003	Determination of antimony, iron and selenium in water by electrothermal atomic absorption spectrometric method
11917 : 2018/ ISO 8968-1 : 2014	Milk and milk products — Determination of nitrogen content — Kjeldahl principle and crude protein calculation (<i>first revision</i>)	16069 (Part 2) : 2013/ ISO 21527-2 : 2008	Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of yeasts and moulds: Part 2 Colony count technique in products with water activity less than or equal to 0.95
12074 : 1987	Method for determination of lead by atomic absorption spectrophotometer	16072 : 2012	Determination of moisture content in milk powder and similar products (Routine method)
12756 : 2019/ ISO 2962:2010	Cheese and processed cheese products — Determination of total phosphorus content — Molecular absorption spectrometric method (<i>first revision</i>)	16195 : 2014/ ISO/TS 15495 : 2010	Milk, milk products and infant formulae — Guidelines for the quantitative determination of melamine and cyanuric acid by LC-MS/MS
12759 : 2019/ ISO 8156 : 2005	Dried milk and dried milk products — Determination of insolubility index (<i>first revision</i>)	16424 : 2016/ ISO 7251 : 2005	Microbiology of food and animal feeding stuffs — Horizontal method for the detection and enumeration of presumptive <i>Escherichia coli</i> — Most probable number technique
12760 : 2012/ ISO 8070 : 2007	Milk and milk products — Determination of calcium, sodium, potassium and magnesium contents — Atomic absorption spectrometric method (<i>first revision</i>)	16639 : 2018/ ISO 20633 : 2015	Infant formula and adult nutritionals — Determination of vitamin E and vitamin A by normal phase high performance liquid chromatography
13500 : 1992	Spray dried milk powders scorched particles — Determination	16640 : 2018/ ISO 20634 : 2015	Infant formula and adult nutritionals — Determination of vitamin B ₁₂ by reversed phase high performance liquid chromatography (RP-HPLC)
13688 : 2020	Packaged pasteurized milk — Specification (<i>second revision</i>)		
14433 : 2022	Infant milk substitutes — Specification (<i>second revision</i>)		

IS 1656 : 2022

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
16642 : 2018/ ISO 20639 : 2015	Infant formula and adult nutritionals — Determination of pantothenic acid by ultra high performance liquid chromatography and tandem mass spectrometry method (UHPLC-MS/MS)	17670 : 2021/ ISO 23305 : 2020	Fortified milk powders, infant formula and adult nutritionals — Determination of total biotin by liquid chromatography coupled with immunoaffinity column clean-up extraction
16649 : 2018/ ISO 20637 : 2015	Infant formula and adult nutritionals — Determination of Myo — Inositol by liquid chromatography and pulsed amperometry	IS/ISO 21446 : 2019	Infant formula and adult nutritionals — Determination of trans and total cis trans Vitamin K ₁ content — Normal phase HPLC
17176 : 2019/ ISO 20635 : 2018	Infant formula and adult nutritionals — Determination of vitamin C by (Ultra) high performance liquid chromatography with ultraviolet detection ((U) HPLC-UV)	ISO 14377 : 2002	Canned evaporated milk — Determination of tin content — Method using graphite furnace atomic absorption spectrometry
17177 : 2019/ ISO 20636 : 2018	Infant formula and adult nutritionals — Determination of vitamin D by liquid chromatography mass spectrometry	ISO 15151 : 2018	Milk, milk products, infant formula and adult nutritionals — Determination of minerals and trace elements — Inductively coupled plasma atomic emission spectrometry (ICP-AES) method
17379 : 2020/ ISO 20647 : 2015	Infant formula and adult nutritionals — Determination of total iodine - Inductively coupled plasma mass spectrometry (ICP-MS)	ISO 15213 : 2003	Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of sulfite-reducing bacteria growing under anaerobic conditions
17668 : 2021/ ISO 21468 : 2020	Infant formula and adult nutritionals — Determination of free and total choline and free and total carnitine — Liquid chromatography tandem mass spectrometry (HPLC-MSMS)	ISO 20649 : 2015	Infant formula and adult nutritionals — Determination of chromium, selenium and molybdenum — Inductively coupled plasma mass spectrometry (ICP-MS)
17669 : 2021/ ISO 21470 : 2020	Infant formula and adult nutritionals — Simultaneous determination of total vitamins B ₁ , B ₂ , B ₃ and B ₆ Enzymatic digestion and LC-MSMS	ISO 21424 : 2018	Milk, milk products, infant formula and adult nutritionals — Determination of minerals and trace elements — Inductively coupled plasma mass spectrometry (ICP-MS) method

ANNEX B

[Table 2, SI No. (iii)]

DETERMINATION OF FAT

B-1 APPARATUS

As prescribed in 6 of IS 11721.

B-2 REAGENTS

As prescribed in 5 of IS 11721 and the reagents given in B-2.1 and B-2.2.

B-2.1 Iodine Solution, 0.1 N.

B-2.2 Diastase or Amylase

B-3 PROCEDURE

B-3.1 Weigh accurately about 1 g of the sample in a Mojonnier fat extraction flask. After introducing

the sample into the flask, add approximately 0.5 g diastase or amylase. Then add 8 to 10 ml distilled water (see IS 1070) at 45°C to facilitate suspension.

B-3.2 Place the stoppered extraction flask in the water-bath for 2 hours, shaking it from time to time taking care that the product does not stick to the walls of the flask. Check if the starch is digested. Add 2 drops of iodine solution. No blue colouration should appear. If required, place the flask again in water-bath till the digestion of the starch is complete. Proceed further as prescribed in 9.2 to 10.1 of IS 11721, excluding 9.4.1.

ANNEX C

[Table 2, SI No. (iv)]

DETERMINATION OF TOTAL CARBOHYDRATES

C-1 Total carbohydrates are calculated as follows, after determining the percentage of moisture, total protein, fat and total ash:

Total carbohydrates, including sucrose, dextrose and dextrins, maltose or lactose, percent by mass =

$$100 - (A + B + C + D)$$

where

A = percent by mass of moisture;

B = percent by mass of total protein;

C = percent by mass of fat; and

D = Total ash, percent by mass.

ANNEX D

[Table 3]

SAMPLING PLAN FOR MICROBIOLOGICAL REQUIREMENTS

D-1 SAMPLING PLAN FOR MICROBIOLOGICAL REQUIREMENTS

The terms *n*, *c*, *m* and *M* used in this standard have the following meaning:

n = Number of units comprising a sample;

c = Maximum allowable number of units having microbiological counts above *m* for 2-class sampling plan and between *m* and *M* for 3-class sampling plan;

m = Microbiological limit that separates unsatisfactory from satisfactory in a 2-class sampling plan or acceptable from satisfactory in a 3-class sampling plan; and

M = Microbiological limit that separates unsatisfactory from satisfactory in a 3-class sampling plan.

IS 1656 : 2022

D-2 INTERPRETATION OF RESULTS

2-Class Sampling Plan (where n, c and m are specified)	3-Class Sampling Plan (where n, c, m and M are specified)
1. Satisfactory, if all the values observed are $\leq m$ 2. Unsatisfactory, if one or more of the values observed are $> m$ or more than c values are $> m$	1. Satisfactory, if all the values observed are $\leq m$ 2. Acceptable, if a maximum of c values are between m and M and the rest of the values are observed as $\leq m$ 3. Unsatisfactory, if one or more of the values observed are $> M$ or more than c values are $> m$

ANNEX E

(Foreword)

COMMITTEE COMPOSITION

Dairy Products and Equipment Sectional Committee, FAD 19

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Research Institute, Karnal	DR MANMOHAN SINGH CHAUHAN (Chairman)
All India Food Processors Association, New Delhi	DR K. L. GABA MR VIJAY GAUR (<i>Alternate</i>)
Bihar State Cooperative Milk Producers' Federation Ltd, (COMPED), Patna	MR SUSHIL KUMAR MR RUPESH RAJ (<i>Alternate</i>)
Confederation of Indian Food Trade and Industry, New Delhi	MS VARSHA YADAV DR ANIRUDHA CHHONKAR (<i>Alternate</i>)
Confederation of Indian Industry, New Delhi	MS NEHA AGGARWAL MS ARTI GUPTA (<i>Alternate</i>)
Centre for Analysis and Learning in Livestock and Food (CALF), Anand	DR RAJESH NAIR DR RAJEEV CHAWLA (<i>Alternate</i>)
Directorate of Marketing and Inspection, Faridabad	DR D. M. GOVIND REDDY SHRI RAHUL SINGH (<i>Alternate</i>)
Export Inspection Council of India, New Delhi	DR J. S. REDDY MR KUMAR NARENDER (<i>Alternate</i>)
Envirocare Labs Pvt Ltd, Thane	DR NILESH AMRITKAR DR PRITI AMRITKAR (<i>Alternate</i>)
Food Safety and Standards Authority of India, New Delhi	DR MONICA PUNIYA MS TRIPTI TAYAL (<i>Alternate</i>)
Gujarat Cooperative Milk Marketing Federation Ltd, Anand	MR SAMEER SAXENA MR SAYAN BANERJEE (<i>Alternate</i>)
IDMC Ltd, Anand	MR DEVENDER GUPTA MR PRAKASH MAHESHWARI (<i>Alternate</i>)
Indian Dairy Association, New Delhi	DR G. S. RAJORHIA DR SATISH KULKARNI (<i>Alternate</i>)
Indian Stainless Steel Development Association, Gurugram	MR ROHIT KUMAR MR RAJAT AGGARWAL (<i>Alternate</i>)
Jupitor Glass Works, New Delhi	MR KARAN NANGIA MR AMREEK SINGH PURI (<i>Alternate</i>)
Ministry of Fisheries, Animal Husbandry and Dairying, Department of Animal Husbandry and Dairying, New Delhi	SHRI GOUTAM KUMAR DEB SHRI AJIT KUMAR K. (<i>Alternate</i>)
Mother Dairy Fruit and Vegetable Ltd, Delhi	MS NITA SEN MR SHAILENDER KUMAR (<i>Alternate</i>)
National Dairy Development Board, Anand	MR S. D. JAISINGHANI MR SURESH PAHADIA (<i>Alternate</i>)
National Dairy Research Institute, Karnal	DR VIVEK SHARMA DR RAJESH KUMAR BAJAJ (<i>Alternate</i>)
National Institute of Food Technology Entrepreneurship and Management (NIFTEM), Sonapat	DR P. K. NEMA

IS 1656 : 2022

<i>Organization</i>	<i>Representative(s)</i>
National Institute of Nutrition, Hyderabad	DR B. SANTOSH KUMAR DR SYLVIA FERNANDEZ RAO (<i>Alternate</i>)
Punjab State Coop Milk Producers' Federation Limited	DR SANJEEV KUMAR SHARMA
Rajasthan Co-op Dairy Federation (RCDF) Ltd, Jaipur	MR J. D. SINGH
Tamil Nadu Co-op Milk Producers' Federation Limited, Chennai	MR S. R. SANKAR MR S. JEYACHANDRAN (<i>Alternate</i>)
Tetra Pak India Pvt Ltd, Pune	MR SHASHIKANT RAMNATH SURUSE MR SAMEER SINGH SUHAIL (<i>Alternate</i>)
Vimta Labs Limited, Hyderabad	DR JAGADEESH KODALI DR MUNI NAGENDRA PRASAD POOLA (<i>Alternate</i>)
BIS Directorate General	SHRIMATI SUNEETI TOTEJA, SCIENTIST 'E' AND HEAD (FAD) [REPRESENTING DIRECTOR GENERAL (<i>Ex-officio</i>)]

Member Secretary

DR BHAWANA
SCIENTIST 'D' (FAD), BIS

Panel for Revision of Indian Standards on Infant Foods, FAD 19-FAD 24 Joint Panel 10

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Central Institute of Agricultural Engineering, Bhopal	DR MANOJ TRIPATHI
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(Continued from second cover)

In the formulation of this standard, due consideration has been given to the provisions of the *Food Safety and Standards Act, 2006* and the *Rules and Regulations* framed thereunder; *Infant Milk Substitutes, Feeding Bottles and Infant Foods (Regulation of Production, Supply and Distribution) Act, 1992* and *Rules 1993* and the *Legal Metrology (Packaged Commodities) Rules, 2011*. However, this standard is subject to the restrictions imposed under these, wherever applicable.

The composition of the Committee responsible for formulation of the standard is given in Annex E.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (*second revision*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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